Sobel X

-1	0	1
-2	0	2
-1	0	1

vertical edges, signed

Laplacian

0	1	0
1	-4	1
0	1	0

edges, all directions

sharpen

0	-1	0
-1	5	-1
0	-1	0

identity - Laplacian

emboss

-2	-1	0
-1	1	1
0	1	2

image + relief